PHOTONCANVAS QUICK START

Setting up PhotonCanvas with an xTool F2 Ultra UV

From unpacking the program to your first marked square: installing, connecting, calibrating, the first job, keeping the program up to date, and staying safe with a 355 nm UV laser.

Read this first: UV support is new and untested

PhotonCanvas was built on the F2 Ultra (blue diode and MOPA fiber). Support for the F2 Ultra UV was added from xTool's own published description of how the UV takes jobs, but **nobody has run PhotonCanvas on a real F2 Ultra UV yet**. Every step here is written from the program as it is; where a step hasn't been tried on a UV, it says so. Start small, at low power, on scrap, and stay with the machine.

You need a PhotonCanvas version newer than 0.1.113 (the **?** at the top right → **About PhotonCanvas** shows yours). Older versions list the UV but can't drive it properly.

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1 Before you start

Have these ready:

☐ The F2 Ultra UV unpacked and set up following xTool's own instructions (lens fitted, power, network). Do xTool's first-time setup in **xTool Studio** first, including any firmware update it offers and its own lens and camera setup.

☐ **xTool Studio installed on the same Windows computer.** PhotonCanvas reads its connection keys from it (step 3 of the welcome window). You keep it installed afterwards.

☐ The computer and the machine on the same Wi-Fi or wired network, or joined by the USB cable.

☐ The PhotonCanvas program: `PhotonCanvas-win-x64.zip` (from whoever gave you PhotonCanvas).

☐ Safety glasses rated for **355 nm** (see the next section).

☐ Scrap material to test on: card, plywood, plastic you know is safe, or anodized aluminium. Not PVC or vinyl.

2 Safety with a UV laser

The F2 Ultra UV is a 5 W, 355 nm ultraviolet laser

- **The beam is near-invisible.** The glow you see on the material is not the beam. Never look at the work area while the laser is on with the lid open.
- **Keep the lid closed** while it runs. If you must watch with the lid open, wear safety glasses rated for 355 nm. Glasses for the blue diode or fiber lasers (455 nm, 1064 nm) do **not** protect against UV.
- In Settings → Machine, leave **Allow starting with the lid open unticked**.
- **Never leave a job running unattended.** Keep a fire extinguisher nearby.
- **Ventilate.** Run the exhaust or purifier. Never laser PVC, vinyl or materials you aren't sure about: many plastics give off poisonous gas.
- **Check every job before pressing start.** The preview is a guide, not a guarantee, and on the UV the program itself is untested.
- To stop at once: **Stop** in PhotonCanvas's job bar, or on the machine a short press of its button pauses and a long press while paused cancels. Know where the machine's power switch is.

PhotonCanvas is free, independent hobby software, not made, endorsed or supported by xTool or Makeblock. It comes with no warranty, and using it may not be covered by your machine's warranty.

3 Install PhotonCanvas

There is no installer. The program is a folder that carries everything it needs.

1. Unzip `PhotonCanvas-win-x64.zip` into a folder you can write to, for example `C:\Users\<you>\Apps\PhotonCanvas`. Avoid `Program Files`: updates need to write into the folder.
2. Open the `PhotonCanvas` folder and run `PhotonCanvas.exe`. If Windows SmartScreen warns about an unknown program, choose **More info → Run anyway**.
3. If Windows Firewall asks, **allow** PhotonCanvas on private networks. It needs this to find the machine.

Your own settings, connection keys, materials and calibration are kept in `C:\Users\<you>\PhotonCanvas`, separate from the program, so updating or moving the program loses nothing.

Opening PhotonCanvas connects to the laser

After the first start, PhotonCanvas looks for the machine and connects by itself every time it opens. Connecting never fires the laser or starts a job.

4 The first start: welcome window

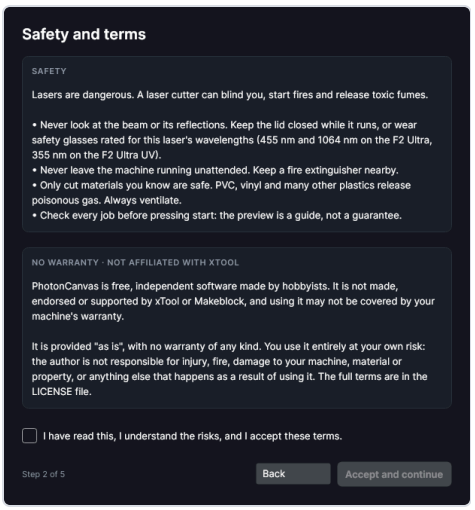
The first time PhotonCanvas opens, the **Welcome to PhotonCanvas** window takes you through five short steps. **Next** moves on and **Back** goes back.

Step 1 · Welcome

What the program is. Click **Next**.

Step 2 · Safety and terms

Read the safety notice and the disclaimer, tick **I have read this, I understand the risks, and I accept these terms.** and click **Accept and continue**. If you close the window without accepting, the program closes.



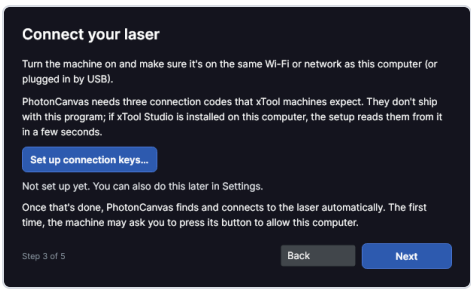
Step 2: the safety notice names 355 nm for the UV.

Step 3 · Connect your laser

xTool machines expect three connection codes. They don't come with PhotonCanvas; it finds them in your copy of xTool Studio.

1. Click **Set up connection keys…**
2. Click **Find them in my xTool Studio**. The keys are read in a few seconds. (You can also choose xTool Studio's program file yourself, or paste the keys.)
3. The step then says ✓ **Connection keys are set up.**

You can skip this and do it later in Settings → Connection.



Step 3: one-time connection keys.

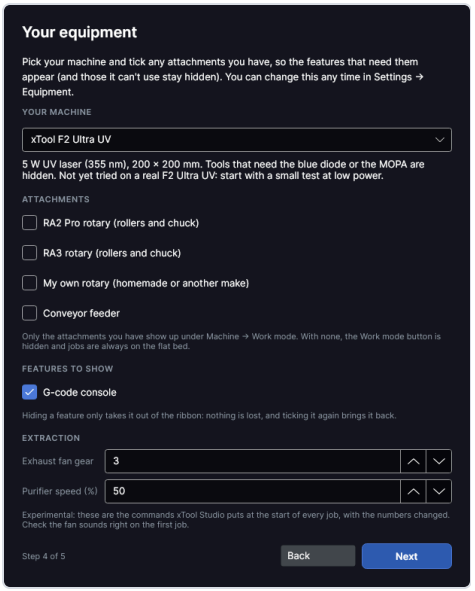
Step 4 · Your equipment: choose the F2 Ultra UV

This is the most important step for a UV owner. Under **YOUR MACHINE**, open the list and choose **xTool F2 Ultra UV** (or **xTool F2 Ultra UV (Class 1)** if you have the enclosed Class 1 model; both take jobs the same way).

With the UV chosen, PhotonCanvas:

- sets the work area to **200 × 200 mm**;
- sends every layer to the UV laser, with a frequency of 40 to 150 kHz;
- hides the tools that need the blue diode or the MOPA fiber laser (colour on metal, diffraction gratings, flip pictures, laser alignment, the diode/MOPA switch, overheat and galvo-timing settings, and the xTool material presets).

Tick an attachment only if you have one. Leave the extraction numbers as they are unless you know your fan settings.



Step 4 with the F2 Ultra UV chosen.

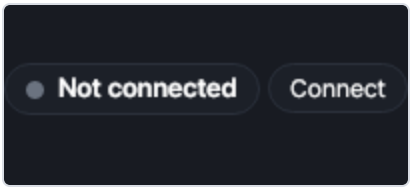
Step 5 · How a job works

A short summary. Click **Start designing**.

You can change the machine later in Settings → **Equipment** (the same list) or Settings → **Machines**.

5 Connect to the machine

1. Turn the F2 Ultra UV on and wait for it to finish starting up.
2. Look at the right end of the ribbon. PhotonCanvas tries to connect by itself; if it shows **Not connected**, click **Connect**.
3. **The first time, the machine may ask you to allow this computer:** press the button on the machine when it lights up or beeps.
4. When connected, the status shows the machine's name and state instead of **Not connected**, and the **Auto-focus** button appears beside it.



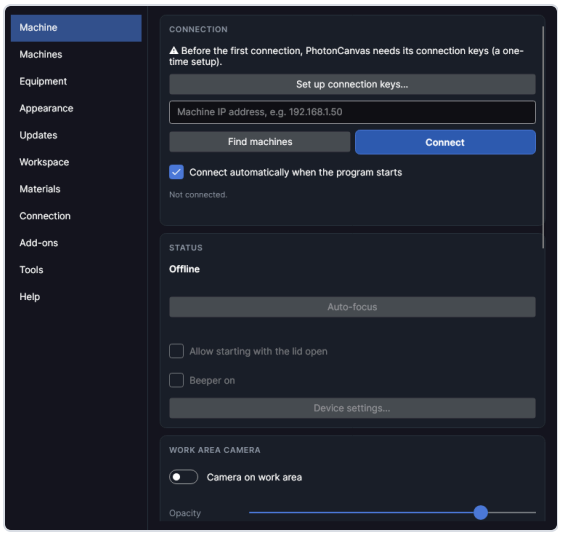
The connection status at the right end of the ribbon, before connecting.

If it doesn't find the machine

Open Settings (the gear at the top right, or Machine → **Machine settings**) and go to the **Machine** page:

- **Find machines** searches the network again.
- Or type the machine's IP address (xTool Studio or your router shows it) and click **Connect**.
- Keep **Connect automatically when the program starts** ticked.
- If it says the connection keys are missing, click **Set up connection keys...** (step 3 above).

Close xTool Studio while PhotonCanvas is connected: if another program is connected, the machine can refuse PhotonCanvas.



Settings → Machine: connection, status, auto-focus.

> **Untested on the UV**
>
> The UV uses the same connection protocol and keys as the F2 Ultra according to xTool's own software, but this hasn't been tried.

Check the machine choice

Settings → **Machines** lists every machine PhotonCanvas knows. Make sure **xTool F2 Ultra UV** is the one selected. If the F2 Ultra is selected instead, jobs would be written for the blue diode, which the UV doesn't have.

Another quick check: in the layer settings panel, a UV layer says **UV laser · 355 nm** and has a **Frequency** box, with no diode/MOPA switch.



Settings → Machines with the F2 Ultra UV selected.

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6 Calibrate

Do these in order. The first is done in xTool Studio; the rest in PhotonCanvas.

6.1 xTool's own UV setup (in xTool Studio)

The F2 Ultra UV's lens and galvo calibration, its small-lens height calibration and firmware updates are xTool's, and PhotonCanvas doesn't do them. Do them in xTool Studio as xTool's manual describes, then close xTool Studio before connecting PhotonCanvas. The calibration is kept on the machine, so PhotonCanvas uses it too.

> **Use the standard lens**
>
> PhotonCanvas works with the UV's standard 200 × 200 mm field. It can't choose the small lens yet, so keep the standard lens fitted.

6.2 Focus

Every job needs the material in focus. Put the material under the head, then click **Auto-focus** at the right end of the ribbon (or on Settings → Machine). The machine measures the material and moves to the right height. Out of focus, marks come out faint or wide.

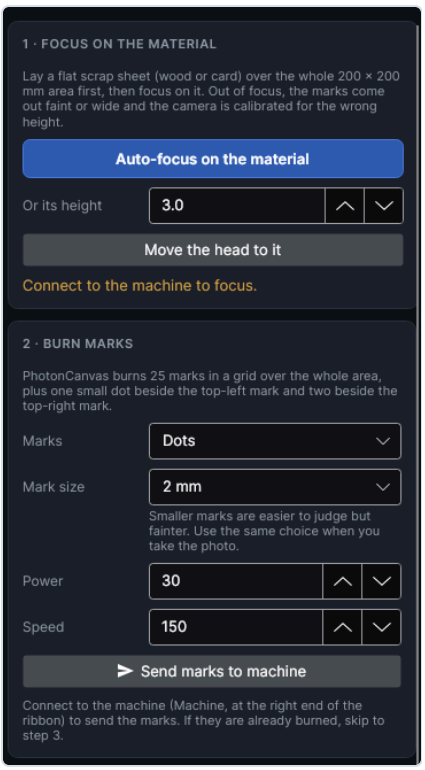
6.3 Calibrate the camera

This lines up the camera picture with where the laser actually marks, so you can place designs on the photo of your material. Machine → **Calibrate camera** opens a window with five steps:

1. **Focus on the material.** Lay a flat scrap (card or thin wood) over the whole 200 × 200 mm area and click **Auto-focus on the material**.
2. **Burn marks.** PhotonCanvas burns 25 small marks in a grid over the whole area (plus three small dots that show which way round it is). Keep the suggested power and speed for a first try; on the UV they're burned with the UV laser at its default frequency. Close the lid and press the machine's start button.
3. **Photo.** Take a photo of the marks with the machine's camera.
4. **Locate the marks.** PhotonCanvas finds them in the photo; follow the window's instructions for any it can't find.
5. **Check and save.** PhotonCanvas works out the correction; save it.

Repeat only if you move the camera or the photo stops lining up.

> **Untested on the UV**
> If the marks are too faint to find, raise the power a little and burn them again on a fresh scrap.



Calibrate camera, steps 1 and 2.

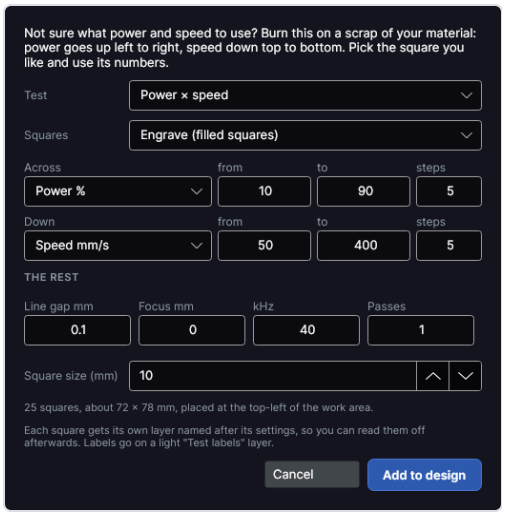
6.4 Find the right settings with a test grid

Each material needs its own power, speed and frequency. A test grid burns a set of small squares, each with different numbers, so you can pick the one you like.

1. Materials & Tests → Tests → **Test grid**.
2. Choose **Power × speed**. The UV starts at **40 kHz** in **kHz**.
3. Click **Add to design**. The squares are added at the top-left of the work area, each on its own labelled layer.
4. Send it like any job (section 7), then read the best square's numbers off its layer name.

Once you have power and speed, run a second grid: choose **Any two settings** and put **Frequency kHz (MOPA, UV)** across (40 to 150 kHz): a higher frequency puts the pulses closer together for smoother, gentler marks.

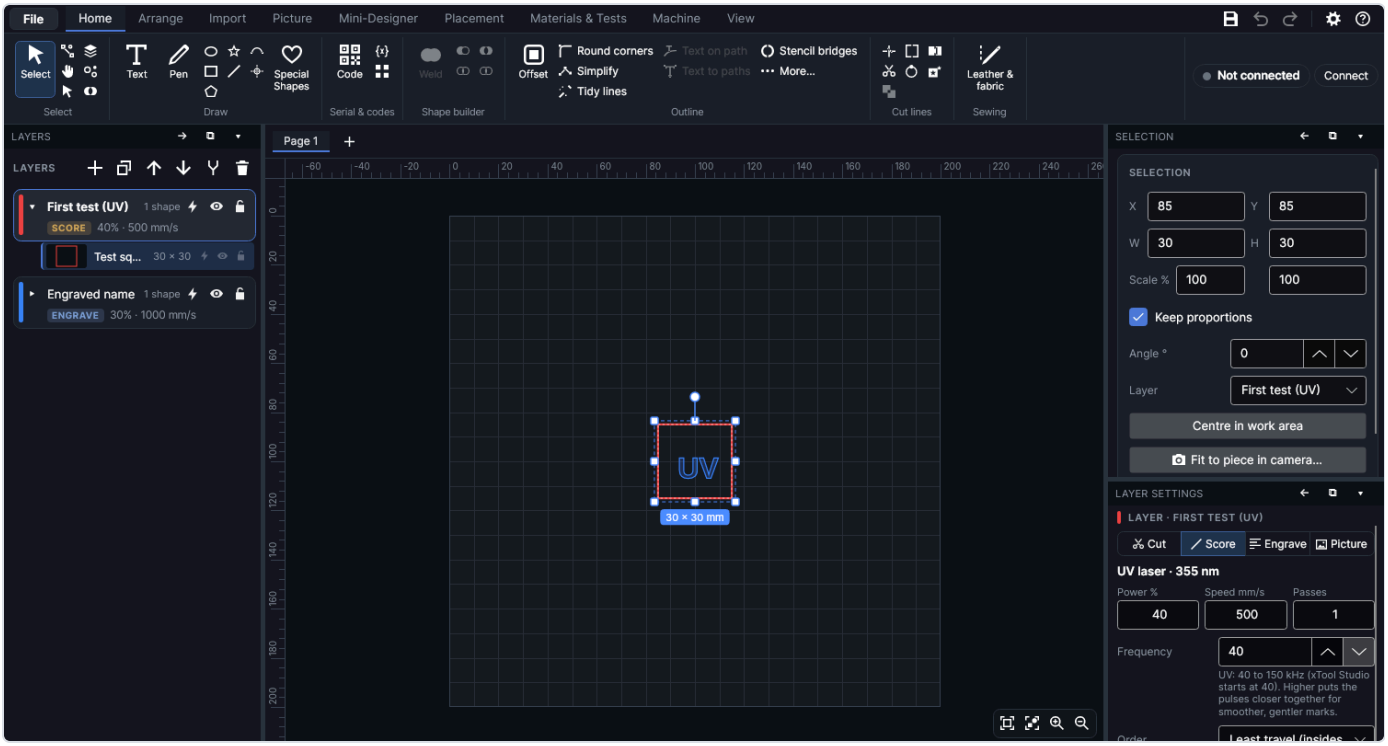
Keep what works in Materials & Tests → **Library**. xTool's own material presets are for the diode and MOPA, so they're hidden on the UV.



The test grid on the UV: 25 squares, 40 kHz.

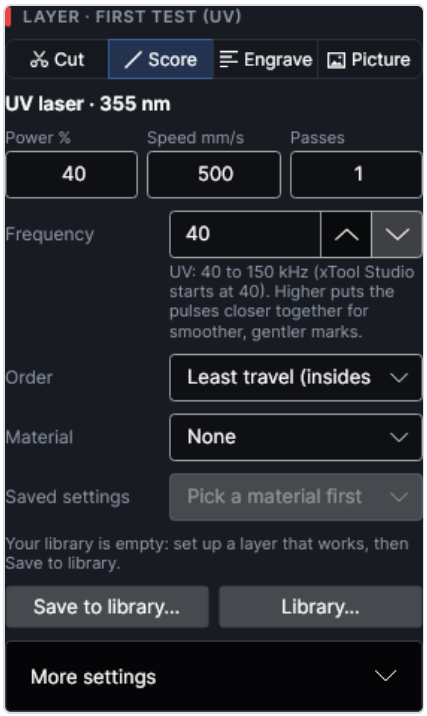
7 Your first job

A 30 mm square outline with "UV" engraved inside it, on scrap card or wood. The numbers below are a cautious **starting guess**, not a tested UV recipe: use your test grid results once you have them.



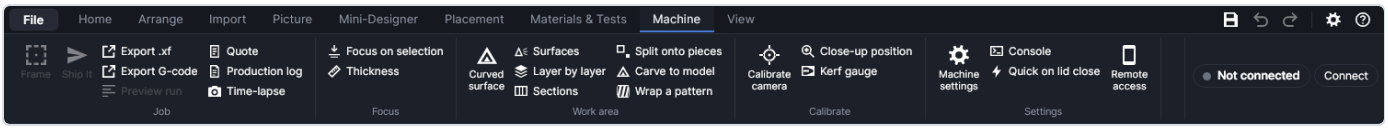
The first job: a square on a Score layer, the letters on an Engrave layer, and the UV layer settings on the right.

1. **Connect** (section 5). The status at the right end of the ribbon must show the machine.
2. **Draw the square.** On the **Home** tab, click the rectangle and drag on the work area. In the **Selection** panel on the right, type **30** in **W** (with **Keep proportions** ticked, **H** follows), then click **Centre in work area**. The grid square is the UV's 200 × 200 mm area; nothing outside it can be sent.
3. **Choose what the laser does.** Each shape sits on a layer (the **Layers** panel on the left). Select the layer and, in **Layer settings**, click **Score**: it marks along the outline without cutting.
4. **Set the numbers: Power %** 40, **Speed mm/s** 500, **Passes** 1, **Frequency** 40. The panel should say **UV laser · 355 nm**.
5. **Add the text** (optional). Click **Text**, click inside the square where the text should start, and type **UV** (the typing goes into the text box in the Selection panel). Add a new layer (**Add layer**, the **+** in the Layers panel), set it to **Engrave** at 30%, 1000 mm/s and 40 kHz, and move the text onto it with the **Layer** box in the Selection panel.



Layer settings on the UV: no diode/MOPA switch, a Frequency box from 40 to 150 kHz.

6. **Place the material and focus.** Lay the scrap in the middle of the bed and click **Auto-focus**.
7. **Frame.** Click **Frame** (in the job bar under the work area, or Machine → **Frame**). The first time, confirm with **Start framing**. The head traces round the job with a faint pointer, over and over. Move the material until the outline sits where you want it, then click **Stop framing**.
8. **Ship It.** Click **Ship It** (job bar, or Machine → **Ship It**). A summary lists each layer as *UV 40 kHz* with its power and speed, the size, the time and anything worth checking. Read it, then click **Ship It** in it to upload. Uploading never starts the laser.
9. **Start.** The job bar says **Uploaded. Press the start button on the machine to begin**. Close the lid, put on your glasses if you'll be near the machine, and press the machine's start button.
10. **Watch it.** The job bar follows the job. **Pause** and **Stop** are there while it runs. Stay with the machine until it finishes.



The Machine tab: Frame and Ship It on the left (they light up once connected), Calibrate camera in the middle, Machine settings on the right.

> **If the mark is too light or too dark**
>
> Too light or patchy: focus again, then raise the power or lower the speed a step. Too dark or melted: lower the power or raise the speed, or try a higher frequency. Change one thing at a time, and use a test grid (6.4) to find good numbers quickly.

8 Install an update

New versions are published from time to time. Updating never touches your settings, keys, calibration or designs.

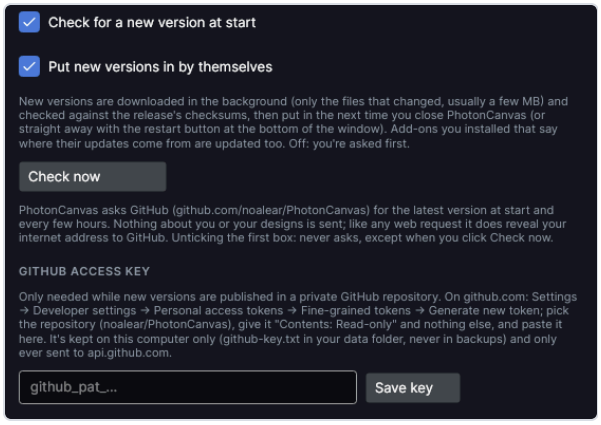
Automatic updates (recommended)

PhotonCanvas's updates are published in a private GitHub repository, so the program needs a **GitHub access key** to see them. Ask the person who gave you PhotonCanvas for one (a read-only key for this program only), then:

1. Open Settings → **Updates**.
2. Paste the key into **GITHUB ACCESS KEY** and click **Save key**. It's kept on this computer only.
3. Leave **Check for a new version at start** and **Put new versions in by themselves** ticked.

From then on, new versions download in the background and go in the next time you close PhotonCanvas (or straight away with the restart button at the bottom of the window). **Check now** looks at once.

With **Put new versions in by themselves** unticked, a window tells you about a new version at start, with **Download**, **Skip this version** and **Later**.



Settings → Updates, with the place for the GitHub access key.

Updating by hand

1. Get the new `PhotonCanvas-win-x64.zip`.
2. Close PhotonCanvas.
3. Unzip it over your existing PhotonCanvas folder, replacing the old files (or into a fresh folder, and delete the old one).
4. Start PhotonCanvas. The **?** at the top right → **About PhotonCanvas** shows the new version.

Nothing of yours is in the program folder (it's all in `C:\Users\<you>\PhotonCanvas`), so nothing is lost.

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9 What doesn't work on the UV yet

PhotonCanvas covers flat marking, engraving and pictures on the UV. These things are not supported yet:

Feature	On the F2 Ultra UV today
Running on a real UV	Not tried yet. Everything UV is written from xTool's published job format.
Small lens and its height calibration	Not supported. Keep the standard 200 × 200 mm lens fitted; do lens setup in xTool Studio.
Inner engraving in glass or crystal (subsurface)	Not supported.
Galvo timing (per-lens delays)	Hidden on the UV; the machine's own settings from xTool Studio are used.
Riser, glass platform, sphere and cylinder modes	Not supported. The rotary attachments are written for the F2 Ultra and haven't been tried on the UV.
Colour on metal, diffraction gratings, flip pictures	Hidden: they need the MOPA fiber laser.
Align lasers, graphene tests	Hidden: they need the blue diode (and MOPA).
xTool material presets	Hidden: they're for the diode and MOPA. Your own Library entries still work.
Camera calibration	Available, but not yet tried on the UV's camera.

Designs made for an F2 Ultra open fine: when the UV is chosen, every layer runs on the UV laser and its MOPA pulse width is ignored. A frequency outside 40 to 150 kHz is refused when you send the job, with a message saying so.

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10 If something goes wrong

What you see	What to do
Status stays **Not connected**	Check the machine is on and on the same network (or USB). Close xTool Studio. Allow PhotonCanvas through Windows Firewall. Settings → Machine → **Find machines**, or type the IP address.
"Connection keys" message	Settings → Machine → **Set up connection keys…** → **Find them in my xTool Studio**. xTool Studio must be installed.

Work area is 220 × 220 mm, or layers show a diode/MOPA switch	The F2 Ultra is chosen, not the UV. Settings → Machines → **xTool F2 Ultra UV**.
"UV frequency" error when sending	Set the layer's Frequency between 40 and 150 kHz.
Something is outside the work area	The UV's area is 200 × 200 mm. Move or shrink the design until it's inside the grid.
Uploaded, but nothing happens	That's by design: press the start button on the machine, with the lid closed.
Marks faint, wide or uneven	Auto-focus again. Check the material lies flat. Then adjust power and speed with a test grid.
Camera picture doesn't line up	Machine → **Calibrate camera** again (6.3).
Anything else	The **?** at the top right → **Help guide** (or F1) opens the full guide that comes with the program. Tell the person who gave you PhotonCanvas what happened, especially anything UV-specific: you're among the first to use it on a UV.